

ESP32-S3 E-Ink E-Reader

User Manual



A minimalist, self-hosted e-reader for plain-text books.

Runs on a LILYGO T5 4.7" E-Paper S3 - Firmware v0.1.0

Upload books over Wi-Fi - Read with automatic bookmarks - Drive it all from one button

1. Overview

This device is a small, self-hosted e-reader for plain-text (.txt) books. You upload books to it over its own Wi-Fi network from a phone or laptop, then read them on the crisp 4.7" e-ink screen. It remembers where you stopped and lets you drop bookmarks - all from a single button.

What you need

- The e-reader board (LILYGO T5 4.7" E-Paper S3).
- A USB-C cable to charge and flash the device.
- Optional: a 1000 mAh Li-Po battery for cordless reading.
- Optional: a microSD card formatted FAT32 for a larger library.
- Any phone or laptop with Wi-Fi and a web browser.

2. Getting Started

- 1 Power the device using the USB-C cable or a charged battery.
- 2 On boot it briefly shows "Starting up...", then opens your last book, or the Library screen if none is saved.
- 3 Books are stored on the microSD card when one is inserted (FAT32); otherwise they are kept on the internal flash memory.

No microSD card required. Without a card the reader automatically falls back to internal storage, so everything still works - you just have less room for books.

3. Adding Books Over Wi-Fi

The reader hosts its own Wi-Fi network and a small web page you use to upload books. No home router or internet is needed.

- 1 On your phone or laptop, open Wi-Fi settings and join the network named EReader-Setup.

Password:

read1234

- 2 Open a web browser and go to the portal address:

`http://ereader.local` (or `http://192.168.4.1`)

- 3 Drag and drop your .txt files onto the page, or click to browse and select them.
- 4 Uploaded books appear in the Library on the device. You can also delete books from the same page.

To save battery, the Wi-Fi portal automatically turns off about 10 minutes after boot. You can switch it back on at any time from the reader's menu or the Library's Wi-Fi row (see Navigation).

4. Navigation - the single button

Everything is controlled with the one on-board button. Rather than counting taps, the reader senses how long you hold it: a quick tap does one thing, a longer hold does another. A brief full-screen flash confirms each action.

While reading a book

GESTURE	WHAT IT DOES
Quick tap	Next page
Medium hold (~0.35s)	Previous page
Long hold (~0.75s)	Open the menu

In the menu or the Library list

GESTURE	WHAT IT DOES
Quick tap	Move the highlight to the next item
Long hold	Select / open the highlighted item

The menu lists Resume reading, Library, Bookmark here, and Wi-Fi upload on/off. The Library shows your books with a Wi-Fi upload toggle as the first row, so uploads are always reachable.

Previous page is a short hold, not a double tap: the e-ink screen takes about 0.8 s to refresh a page, which is too slow to catch a fast double tap reliably, so a brief hold is used instead.

Optional dedicated buttons

If you wire external Previous / Next buttons to the free GPIO pins (see the hardware notes), they turn pages directly without needing the button gestures.

5. Features

- Reads plain-text .txt books with automatic word-wrapped pages.
- Resumes automatically at your last book and reading position.
- Named bookmarks you can drop with a long press.
- On-device Library listing of all uploaded books.
- Built-in Wi-Fi upload portal - no app or cloud account needed.
- Optional USB drive mode: mount the microSD over USB to copy books via cable (separate firmware variant).
- E-ink friendly: periodic full refresh to clear ghosting, and light sleep to save power.

6. Turning Wi-Fi On and Off

Open the menu (long-hold while reading) and select "Wi-Fi upload", or hold on the Wi-Fi row at the top of the Library, to toggle the upload portal. The screen confirms with "Wi-Fi On" (and the join details) or "Wi-Fi Off". Toggling by hand overrides the automatic power-saving shutoff, so it stays exactly as you set it.

7. Power & Battery

- After about 2 minutes idle the reader enters light sleep; press BOOT to wake it.
- Reading works fully offline - Wi-Fi is only needed to add books.

8. Troubleshooting

- "No books yet": upload a .txt via the Wi-Fi portal.
- microSD not detected: make sure it is FAT32 and fully seated; the reader still runs on internal storage without it.
- Can't open ereader.local: use <http://192.168.4.1> instead.
- Portal not showing: open the menu (or the Library's Wi-Fi row) and switch Wi-Fi back on.